F20/21 DL MACHINE LEARNING

GROUP 13

MACHINE LEARNING ON PLAYER-GENERATED GAME DATA

Skill, Strategy and Sketch Recognition



PROJECT PILLARS

Three forms of player-generated game data: behaviour, strategy and visual creation.



Behaviour & Skill

Predict player skill from gameplay behaviour.
Uses SkillCraft features such as actions, latency and hotkey usage to classify skill level.

Ref: Thompson et al. (2013)



Strategy & Outcomes

Predict match outcomes from team composition by using Dota 2 hero selections and match settings to learn strategic patterns linked to winning.

Ref: Wong, Kwok and Tan (2022)



Visual Creation

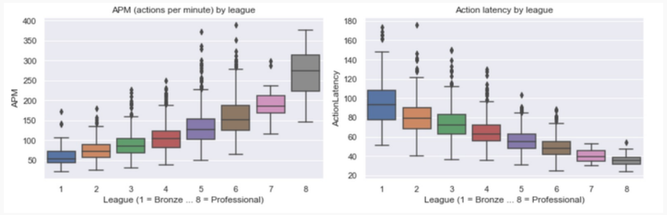
Recognise player-created sketches using CNNs.
Uses Quick, Draw! images to classify what players intended to draw. google quick draw

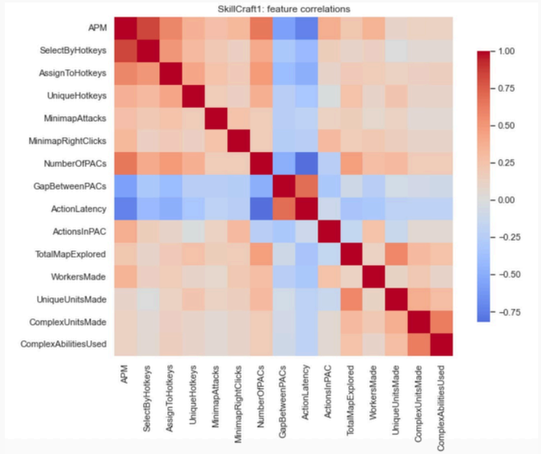
Ref: Ha and Eck (2017)

DATA SPECIFICATIONS

DATASETS & PREPROCESSING

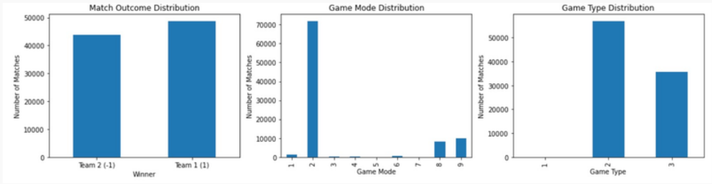
SkillCraft1



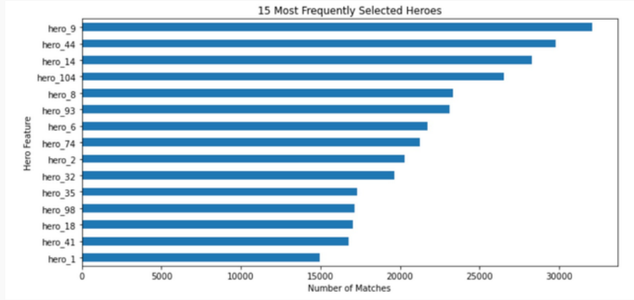


https://archive.ics.uci.edu/dataset/272/skillcraft1%2Bmaster%2Btable%2Bdataset

dota2Test

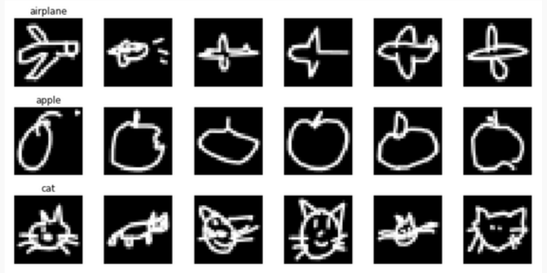


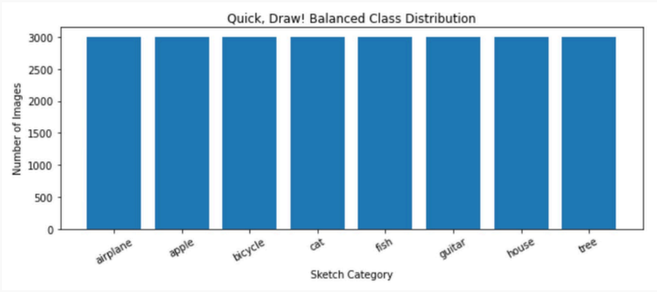
Your paragraph text



https://archive.ics.uci.edu/dataset/367/dota2%2Bgames%2Bresults

QuickDraw





https://github.com/googlecreativelab/quickdraw-datasetragraph text

EXECUTION STRATEGY

WORK PLAN & MOSCOW REQUIREMENTS

MUST HAVE (MVP Backbone)

- Select and justify suitable datasets that match the chosen project topic.
- Clean and preprocess the dataset so it is ready for machine learning.
- Perform exploratory data analysis using graphs and basic statistics
- Train at least three baseline machine learning algorithms and compare their performance.
- Implement and evaluate the required neural network model, such as an NLP or CNN where appropriate.

Should Have (Optimization)

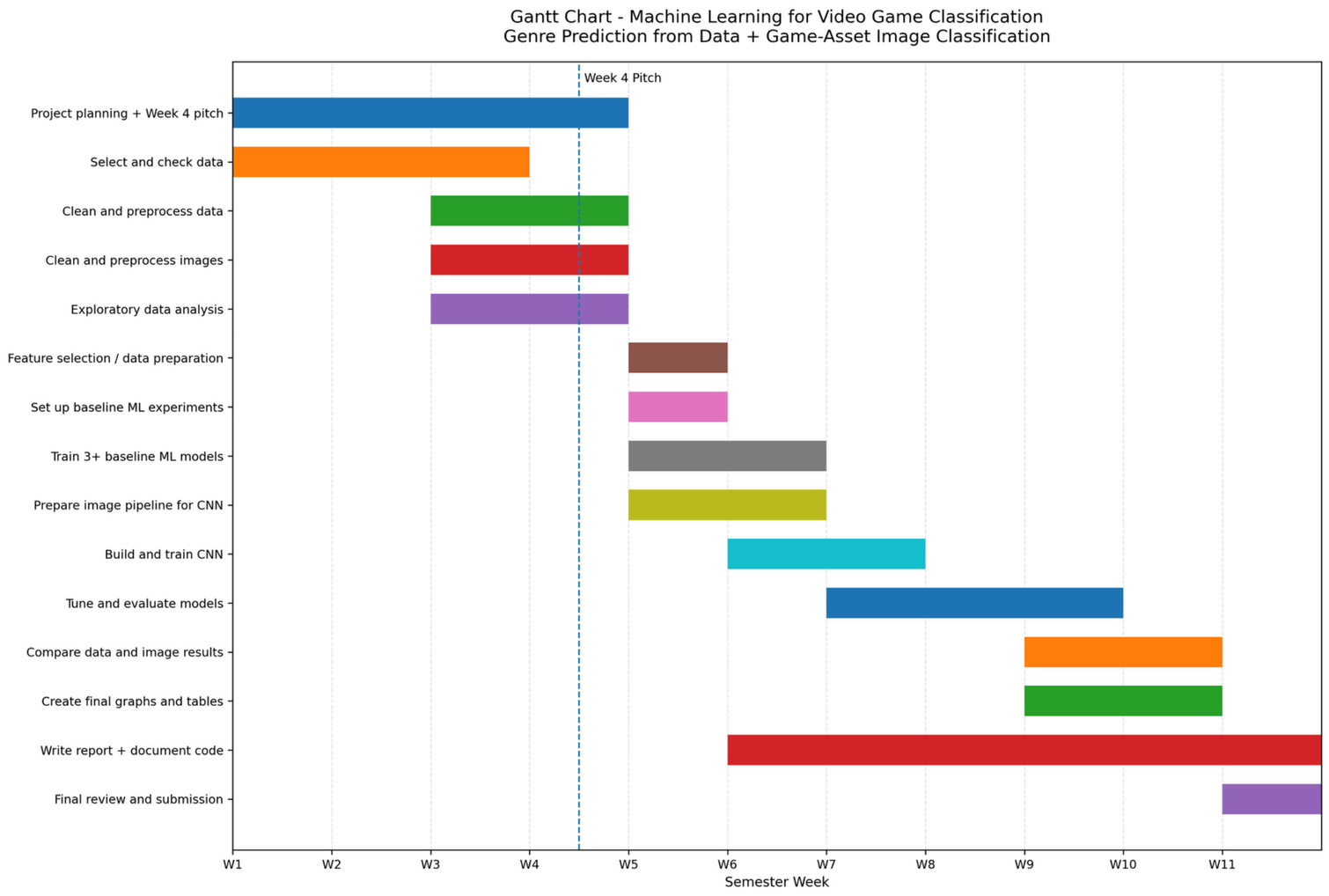
- Perform feature selection or feature evaluation to identify the most useful inputs.
- Improve the models by adjusting their settings.
- Compare the models using suitable evaluation measures

COULD & WON'T HAVE

- **Could Have:** Test additional models, extra visualisations, clustering, or data augmentation if time allows.
- **Won’t Have:** Use generative AI models, unnecessarily complex methods, or other techniques outside the scope of the coursework.

PROJECT ROADMAP

10-WEEK TIMELINE (GANTT SCHEDULED)



FEASIBILITY & LITERATURE

RISK REGISTER & SUPPORTING REFERENCES

RISK ANALYSIS – RISK REGISTER

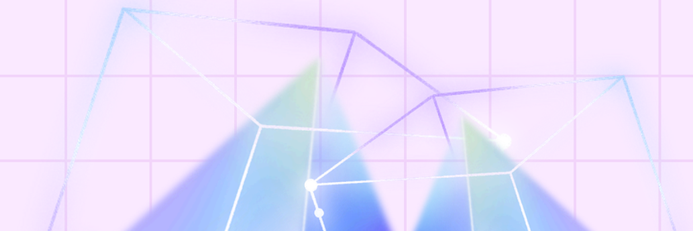
ID	Risk description	Category	Likelihood	Impact	Score (L x I)	Priority
R01	Project falls behind schedule	Project	3	4	12	High
R02	Dataset has missing or poor-quality data	Data	3	3	9	High
R03	Machine learning models perform poorly	Model	3	3	9	High
R04	Project misses coursework requirements	Project	2	4	8	Medium
R05	Dataset is too large to process easily	Technical	2	3	6	Medium
R06	Neural network training takes too long	Technical	2	3	6	Medium
R07	Team member misses assigned tasks	Project	2	3	6	Medium
R08	Code is lost or versions conflict	Technical	2	3	6	Medium

Priority: 1–4 Low · 5–8 Medium · 9–12 High · 13–16 Critical

RISK MITIGATION

ID	Potential risk	Priority	Our response
R01	Project falls behind schedule	High	Set weekly milestones; track Gantt chart; prioritise Must Have tasks.
R02	Dataset has missing or poor-quality data	High	Check, clean and validate the data before modelling.
R03	Machine learning models perform poorly	High	Compare different models; improve preprocessing and tune settings.
R04	Project misses coursework requirements	Medium	Check progress against coursework requirements R1–R4.
R05	Dataset is too large to process easily	Medium	Resize images and use representative samples where suitable.
R06	Neural network training takes too long	Medium	Use smaller models and images; tune training settings.
R07	Team member misses assigned tasks	Medium	Agree on owners and deadlines; redistribute delayed tasks.
R08	Code is lost or versions conflict	Medium	Use GitHub, commit often and keep working backups.

Focus: deliver the Must Have tasks first, then complete optional improvements if time allows.



ISAIAS BIJU NEIL HARIKRISHNAN STEVE EVANGELISTA RITIK MULANI ARYAN SATGHARE

REFERENCES

(1) HA, D. AND ECK, D. (2017) 'A NEURAL REPRESENTATION OF SKETCH DRAWINGS', ARXIV PREPRINT, ARXIV:1704.03477. AVAILABLE AT: HTTPS://ARXIV.ORG/ABS/1704.03477 (ACCESSED: 9 OCTOBER 2026)

(2) THOMPSON, J.J., BLAIR, M.R., CHEN, L. AND HENREY, A.J. (2013) 'VIDEO GAME TELEMETRY AS A CRITICAL TOOL IN THE STUDY OF COMPLEX SKILL LEARNING', PLOS ONE, 8(9), E75129. AVAILABLE AT: HTTPS://DOI.ORG/10.1371/JOURNAL.PONE.0075129 (ACCESSED: 9 OCTOBER 2026).

(3) WONG, N.H.L., KWOK, S.-I. AND TAN, J.L. (2022) 'PREDICTIVE ANALYTICS OF FIRST BLOOD AND MATCH OUTCOME IN DOTA 2', TENCON 2022 – 2022 IEEE REGION 10 CONFERENCE (TENCON). HONG KONG, 1–4 NOVEMBER. IEEE. AVAILABLE AT: HTTPS://DOI.ORG/10.1109/TENCON55691.2022.9977786 (ACCESSED: 9 OCTOBER 2026)

THANK YOU